Global Optimisation of Polynomial Portfolio Risk: What the Lasserre Hierarchy Certifies

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Abstract

I ask whether a certified global minimiser of a degree-4 portfolio risk polynomial is worth having. I solve the order-2 Lasserre relaxation, dense and with correlative sparsity, and certify a point a posteriori when its value matches the bound, to solver accuracy. On 1760 synthetic instances, the probability that one local start misses the global minimum grows with the number of assets: in the default regime it is 0.30 at $n = 15$ and 0.83 at $n = 50$. The relaxation certifies all but 29 of the instances, and only 74% of the default-regime instances at $n = 50$. On 37 US stocks (2015–2025), the certified minimiser lowers volatility and CVaR relative to equal weights, as any risk-minimising rule does. It is not better than Ledoit–Wolf minimum variance or minimum CVaR on those measures, and no net-return difference is distinguishable from zero. A Frobenius-ball formulation meant to protect against kurtosis estimation error does not beat the nominal problem, except marginally at the largest noise level; shrinkage towards a structured target does better. A first version of this work contained errors, which I corrected before running the experiments. The conclusions hold for one family of synthetic polynomials and one fixed universe.

1 Introduction

Portfolio risk that accounts for skewness and kurtosis is a polynomial in the weights. With the first four moments the objective has degree four and is not convex in general. A local solver returns a stationary point. The Lasserre hierarchy [7] returns a lower bound, and a feasible point whose value is close to that bound is close to the global optimum, to solver accuracy. I ask what this certificate is worth on portfolio risk polynomials.

I do three things. I implement the order-2 relaxation, dense and with correlative sparsity [13], and a certificate that does not depend on a rank test. I compare it with multi-start local search on 1760 synthetic instances (Experiment A). I test a Frobenius-ball formulation for estimation error on the kurtosis coefficients (Experiment B), and the certified minimiser on 37 US stocks (Experiment C).

Three results stand out. Local search fails more often than I expected, even on the default synthetic regime. In the default regime, one start misses the global minimum with probability 0.30 at $n = 15$ and 0.83 at $n = 50$. The certificate is available in most cases, but its cost at $n = 50$ exceeds that of 100 local runs. On real data the certified minimiser differs from the local one on few dates. It brings no out-of-sample gain over the risk-based baselines.

The first version of this work (tag `v1.0-paper`) contained errors. The sparse budget constraint decoupled the cliques, a size-reduction ratio mixed two metrics, a dense timing was unsupported, and a benchmark described as non-convex is convex. Appendix A lists them with the measured values. I fixed them before running any experiment below.

2 Background

This section defines the finance used below. Appendix E gives the formula of each indicator as implemented, Appendix F the baseline strategies and Appendix G the vocabulary.

Portfolio, returns, drift. A portfolio of n assets is a vector of weights $x \in \mathbb{R}^n$, where x_i is the share of wealth held in asset i. The budget constraint $\mathbf{1}^\top x = 1$ means that all wealth is invested. Long-only means $x \geq 0$: no short sales. The simple return of an asset on day t is $r_t = P_t/P_{t-1} - 1$; the log return is $\log(P_t/P_{t-1})$. P_t is the adjusted close, corrected for splits and dividends. A portfolio earns $x^\top r_t$ in simple returns; log returns do not add up across assets. The experiments estimate on daily log returns and evaluate on simple returns, as the code does. Between two rebalancing dates the weights drift: $x_i \leftarrow x_i(1 + r_{t,i})/(1 + x^\top r_t)$. Rebalancing is the trade back to the target weights, and it costs money in proportion to what is traded.

Variance, covariance, correlation. With T daily returns, Σ_{ij} is the sample covariance of assets i and j, and $x^\top \Sigma x$ is the variance of the portfolio return. Volatility is its square root; it is annualised by $\sqrt{252}$, the number of trading days in a year. The correlation $\Sigma_{ij}/\sqrt{\Sigma_{ii}\Sigma_{jj}}$ lies in $[-1, 1]$. The median pairwise correlation of the 37 stocks is 0.38: diversification lowers risk but does not remove it.

Markowitz. Minimum variance solves $\min_x x^\top \Sigma x$ subject to $\mathbf{1}^\top x = 1$. Without sign constraints the solution is $\Sigma^{-1}\mathbf{1}/(\mathbf{1}^\top \Sigma^{-1}\mathbf{1})$. With $x \geq 0$ the problem is a convex quadratic program, and any solver returns its global optimum. Adding a target for the mean return traces the efficient frontier. Variance does not see asymmetry or tail weight: two portfolios with the same variance can lose very different amounts on their worst days.

Skewness and kurtosis. For a return p with mean μ and standard deviation s, skewness is $E[(p-\mu)^3]/s^3$ and kurtosis is $E[(p-\mu)^4]/s^4$. A normal law has skewness 0 and kurtosis 3. On the 37 stocks (4 172 daily log returns) the median skewness is -0.24, from -1.28 to 0.96, and it is negative for 31 stocks. The median kurtosis is 16.1, from 7.1 to 31.1: extreme days are much more frequent than a normal law predicts (fat tails).

Why a polynomial of degree four. For $p = x^\top r$ the moment of order k is $E[p^k] = \sum_{i_1,\ldots,i_k} x_{i_1} \cdots x_{i_k} E[r_{i_1} \cdots r_{i_k}]$. It is a homogeneous polynomial of degree k in x, and its coefficients are the co-moments of the returns. Variance has degree two, skewness three, kurtosis four. A risk function that penalises variance and kurtosis and rewards skewness is therefore a polynomial of degree four. A fourth-order expansion of expected utility is the usual argument for these three terms. The full fourth co-moment has 91 390 distinct coefficients for 37 assets, against 27 972 observations in a 756-day window. This is why Experiment C also uses a sector model with far fewer coefficients.

Three sources of error. *Model error*: the polynomial is not the investor's true objective (here it has no return term). *Estimation error*: its coefficients are estimated from a finite window. *Optimisation error*: the solver returns a point whose value exceeds the minimum of the polynomial. The certificate controls only the third. It bounds $f(\hat{x}) - \min f$ for the polynomial as given, and says nothing about the first two. Experiment A studies the third on synthetic polynomials, Experiment B the second by drawing errors on the coefficients, and Experiment C mixes all three.

3 Risk model

The synthetic objective is

$$f(x) = x^\top \Sigma x + \sum_i \eta_i x_i^3 + \sum_{i,j} \kappa_{ij}\, x_i^2 x_j^2 + \text{concentration terms}, \tag{1}$$

Table 1: Notation.

Symbol	Financial meaning	Mathematical meaning
n, x	number of assets; portfolio weights	number of variables; point of $\mathbb{R}^n$
r_t, Σ	daily asset returns; their covariance	vector in $\mathbb{R}^n$; positive semidefinite matrix
f, $f^\star$	risk to minimise; lowest risk attainable	polynomial of degree four; its global minimum on $\mathcal{X}$
$\mathcal{X}$	admissible portfolios (long-only, budget)	polyhedron $\{x \geq 0,\ b_{\mathrm{lo}} \leq \mathbf{1}^\top x \leq b_{\mathrm{hi}}\}$
K	number of local starts in Experiment A	number of random SLSQP starts, from 1 to 100
b_{lo}, b_{hi}	least and greatest invested share (synthetic markets)	bounds on $\mathbf{1}^\top x$
η_i, κ_{ij}	skewness-like and kurtosis-like coefficients of the synthetic model	coefficients of x_i^3 and $x_i^2 x_j^2$; not co-moments
d, y_α	—	relaxation order; variable standing for the monomial x^α
$M_d(y)$	—	moment matrix, constrained to be positive semidefinite
λ_{lb}	certified lower bound on the risk	optimum of the relaxation, $\lambda_{\mathrm{lb}} \leq f^\star$
$\hat{x}$	portfolio returned	feasible point tested by the certificate
I_k, s_k	sector k; its total weight	clique of the interaction graph; auxiliary variable
$m_k(x)$	sample moment of order k of the centred portfolio return	polynomial of degree k in x
ℓ_k, w_k	scale of each term; weight chosen for it	coefficients of $F = \ell_2 m_2 - \ell_3 m_3 + \ell_4 m_4$
ρ	relative size of the error on the coefficients (Exp. B)	$\|\varepsilon\|_W / \|\theta\|_W$
δ, γ	robustness margin; its relative size	radius of the Frobenius ball; $\gamma = \delta / \|\theta\|_W$
s	shrinkage intensity	weight of the target in $(1-s)\hat{\kappa} + s\,\kappa_{\mathrm{target}}$
N_{eff}	effective number of assets	$1/\sum_i x_i^2$

on $\mathcal{X} = \{x \geq 0,\ b_{\mathrm{lo}} \leq \mathbf{1}^\top x \leq b_{\mathrm{hi}}\}$ with $b_{\mathrm{lo}} = 0.95$ and $b_{\mathrm{hi}} = 1$. The coefficients η_i play the part of skewness and the κ_{ij} that of kurtosis. The last terms are a stylised concentration penalty on the weights. They are quartic so that f has degree four and they are larger on concentrated positions. They are not calibrated and not derived from an execution model. The coefficients κ_{ij} are model parameters. They are not the fourth co-moment tensor of a portfolio, which would be a dense contraction $\sum \kappa_{ijkl} x_i x_j x_k x_l$. Cross terms exist only within clusters, so the interaction graph is block-structured. The markets are synthetic and uncalibrated in every regime of Experiment A.

On market data (Experiment C) the objective is estimated on a window of 756 days: $F(x) = \ell_2 m_2(x) - \ell_3 m_3(x) + \ell_4 m_4(x)$, with $m_k(x)$ the sample moment of order k of the centred portfolio return $x^\top \tilde{r}_t$. The weights ℓ_k are frozen in advance so that each term contributes the same amount on a single-asset portfolio. There is no return term: the comparison is about risk. A sector variant keeps only intra-sector co-moments, which is a modelling choice and not a property of the data.

4 Relaxations and certificates

The relaxation of order d replaces each monomial x^α with $|\alpha| \leq 2d$ by a variable y_α. It asks that the moment matrix $M_d(y)$ and the localising matrices of the constraints be positive semidefinite, and minimises $\sum_\alpha f_\alpha y_\alpha$. The optimum λ_{lb} satisfies $\lambda_{\mathrm{lb}} \leq f^\star$. Since f has degree four, $d = 2$ is the smallest usable order. Under an Archimedean condition λ_{lb} converges to $f^\star$ as d grows [7, 11]. Convergence at $d = 2$ is not guaranteed.

Correlative sparsity. Let $I_1, \ldots, I_m$ be the maximal cliques of a chordal extension of the interaction graph. The sparse relaxation uses one moment matrix per clique. Waki et al. [13] establish that it is valid and weaker than the dense one. Lasserre [8] proves that, under the running intersection property, it converges to $f^\star$ as the order grows. At equal order, $\lambda_{\mathrm{lb}}^{\mathrm{sparse}} \leq \lambda_{\mathrm{lb}}^{\mathrm{dense}}$. Without the budget variables this is immediate, since each sparse block is a principal submatrix of

a dense block. With the sector-sum variables of the next paragraph, substituting $s_k = \sum_{i \in O_k} x_i$ in a dense moment vector gives a sparse-feasible point with the same objective; a regression test checks it numerically.

The budget $\mathbf{1}^\top x \in [b_{\text{lo}}, b_{\text{hi}}]$ involves every variable and belongs to no clique. Version 1 replaced it by a constraint on $E[\sum x]$, which decoupled the cliques. I now add the sector sums $s_k = \sum_{i \in O_k} x_i$, where O_k is the set of variables owned by clique k. The equalities $s_k = \sum_{i \in O_k} x_i$ live in the clique $I_k \cup \{s_k\}$ and the budget lives in a clique on $\{s_1, \ldots, s_m\}$. This is the star variant. For many cliques I use partial sums $p_k = p_{k-1} + s_k$, which give cliques of size three (chain variant). For disjoint cliques, which are the clusters of this paper, both satisfy the running intersection property. With overlapping cliques the budget clique closes a cycle, the property fails, and the builder warns; the relaxation stays valid.

Certificate. After solving, I take candidate points from the Henrion–Lasserre extraction [6] when the rank condition holds, and from the first-order moments otherwise. I polish them with a local solve and check feasibility to 10^{-8}. A point $\hat{x}$ is certified when $f(\hat{x}) - \lambda_{\text{lb}} \leq \max(10^{-6}, 10^{-4}|\lambda_{\text{lb}}|)$. Since $\lambda_{\text{lb}} \leq f^\star \leq f(\hat{x})$, this holds whatever the rank of the moment matrix. The reported bound is the primal objective of the moment problem, not the dual objective, so the statement holds to solver accuracy (Appendix B). The flat-extension test of Curto and Fialkow [1] is reported as a secondary diagnostic. Solves use CLARABEL [5] through CVXPY [4]. I set its three tolerances (gap absolute, gap relative, feasibility) to 10^{-7}; its defaults are 10^{-8}. The status `optimal_inaccurate` means that only its reduced tolerances were met (gap 5×10^{-5}, feasibility 10^{-4}). The bound is only as accurate as the solver; Appendix B measures the accuracy.

5 Expectations and measurements

Before the experiments I wrote down what I expected, from theory or from the claims of version 1. Table 2 compares these expectations with the measurements. Each line is detailed in the section indicated.

6 Experiments

6.1 A two-asset check

The 2-asset benchmark of version 1 has a positive definite Hessian on $\mathcal{X}$ (smallest eigenvalue 1.68) and a single local minimum. It validates the code but illustrates nothing about local minima. I built a non-convex instance by raising the skewness scale until a multi-start run finds two minima (smallest Hessian eigenvalue -3.46). The best local minimum has value -0.9849 and the other -0.5640; 41% of the local starts end in the worse one. The order-2 bound is -0.9849 and the certificate gap is 3.2×10^{-9}. At $d = 3$ the bound is the same to 10^{-8}, but the rank test fails (ranks 5 and 1). So the certificate is available at $d = 2$ even though the rank test, applied at a higher order, would reject it.

6.2 Experiment A: local search against the relaxation

I generate 1760 instances with $n \in \{2, 4, 6, 8, 10, 15, 30, 50\}$ assets in three regimes, 50 to 120 per cell. A regime fixes the scale of the three families of coefficients of the risk model. In the default regime the skewness scale is 0.3, the kurtosis scale 0.5 and the concentration scale 0.15. The mixed regime raises the skewness scale to 1.0. The stress regime uses skewness scales from 1.0 to 4.0, kurtosis scales from 0.3 to 0.5 and concentration scale 0.05. For each instance I solve the sparse relaxation at $d = 2$ (the dense one for $n \leq 10$) and run 100 SLSQP starts drawn i.i.d. on the

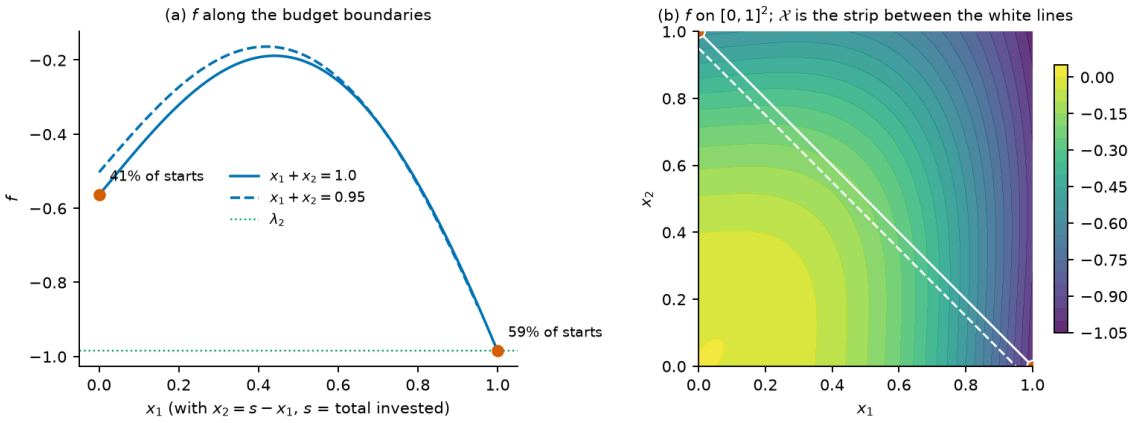


Figure 1: Non-convex 2-asset instance. (a) f along the two budget boundaries; dots are the local minima with the share of starts that reach them. (b) f on the unit square; $\mathcal{X}$ is the strip between the white lines.

budget set. The reference optimum f_{cert} is the certified point when there is one, and the best of 100 runs otherwise. A search with K starts fails when its best value exceeds f_{cert} by more than $\max(10^{-6}, 10^{-4}|f_{\text{cert}}|)$.

Expected. In the default regime I expected local search to find the global minimum, and in the stress regime I expected failures.

Observed. The share of instances with several local minima grows with n. In the default regime it is 0% at $n = 4$, 48% at $n = 15$ and 98% at $n = 50$.

The probability that one start fails grows with n in all regimes (Table 3). In the default regime it is 0.30 at $n = 15$ and 0.83 at $n = 50$. A start fails when it ends above the reference optimum by more than $\max(10^{-6}, 10^{-4}|f_{\text{cert}}|)$. Read 0.83 as: of 100 random starts, about 83 end in a worse point (the mean over the 50 instances of that cell). Lower is better; 0 means that every start finds the optimum.

More starts help without closing the gap (Figure 2). In the default regime at $n = 50$, the best of one start fails in 86% of the instances and the best of 100 in 16.0%.

Reference bias. On an instance that is not certified, the reference is the best of 100 runs, so 100 starts cannot fail there by construction. This concerns 29 instances in total, but 13 of the 50 default-regime instances at $n = 50$. Among the certified ones the 100-start failure rate is 21.6%, not 16.0%. Table 3 gives both rates by regime and n. Rates pooled over all instances depend on the design (eight sizes, three regimes, unequal numbers of seeds), so I do not report them.

Reading. The default regime is not an easy regime beyond about ten assets. I did not test why local minima multiply with n.

The relaxation certifies all but 29 of the 1760 instances, but unevenly: 74% in the default regime at $n = 50$. Certified means that a feasible point has a value within $\max(10^{-6}, 10^{-4}|\lambda_{\text{lb}}|)$ of the bound, to solver accuracy (Appendix G). 79 sparse solves return an inaccurate status. I also solved the dense relaxation on 1100 instances. The difference $\lambda_{\text{lb}}^{\text{dense}} - \lambda_{\text{lb}}^{\text{sparse}}$ has median 2.9×10^{-10}. Only 5 instances exceed 10^{-3} (maximum 9.0×10^{-3}). All five are non-certified instances with $n = 4$ or $n = 8$, and all have two cliques. At order 3 the sparse and dense bounds of the 3 four-asset instances agree to 2.9×10^{-7}, so the gap is specific to order 2. The per-clique ball constraint changes the sparse bound by at most 8.8×10^{-8}. I read the gap as the missing cross-clique moments at order 2 (measured for $n = 4$; order 3 was not run for the 8-asset instances). The smallest difference is -1.0×10^{-6}, which I take to be solver accuracy. Across all instances, the sparse bound exceeds the best local value by at most 9.4×10^{-6}, again the order of the solver tolerance.

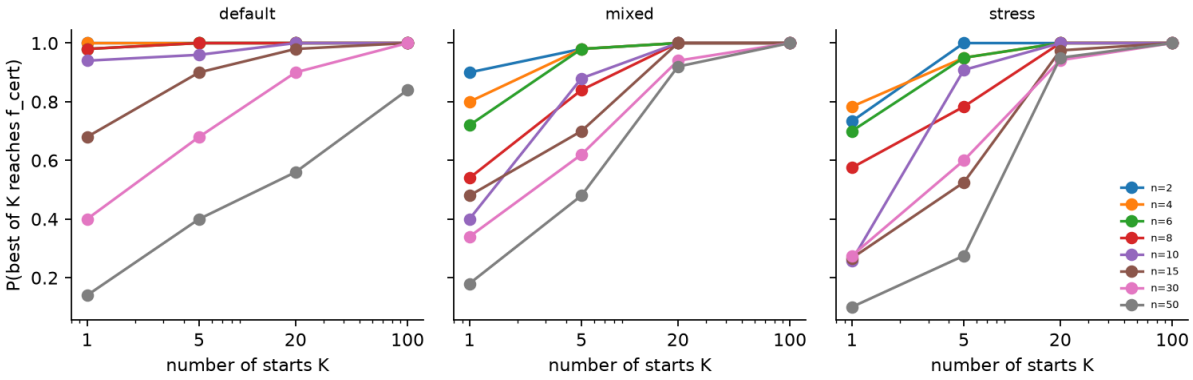


Figure 2: Experiment A. Share of instances where the best of K local starts reaches the reference optimum, by regime and number of assets n.

On time, the sparse relaxation is not cheaper than local search at the largest size. At $n = 50$ the median solve takes 80 s in the default regime against 33 s for 100 local runs, and 20 s against 2 s in the stress regime. These are wall times from six concurrent workers and are indicative only.

6.3 Size and cost of the relaxation

Table 4 gives measured and extrapolated costs. Up to $n = 15$ the dense relaxation is measured: 193 s and 4.7 GB, against 0.71 s for the sparse one on a similar instance. At $n = 12$ and $n = 15$ the dense solve ends with an inaccurate status.

Beyond $n = 15$ the dense figures are extrapolations. The fit has an exponent of 4.35 on the block size and five fitting points. Its interval at $n = 50$ spans 6.2×10^5 to 2.4×10^7 s, and I report it as such.

Take $n = 50$ with ten cliques of five assets. The total number of PSD entries falls by a factor of 128 (star budget) to 215 (chain). The number of free moments falls by 104 to 135. For one instance (seed 0, default parameters, ten cliques of five assets) run alone, the sparse solve takes 26 s (chain) or 51 s (star). In Experiment A the median at $n = 50$ in the default regime is 80 s (star budget, six concurrent workers, build included).

Version 1 reported 398, which is 1 758 276/4 410: the entries of the dense moment matrix over the entries of the ten clique moment matrices. It counted no localising matrix and no budget clique, since it did not impose the budget pointwise. The same moment-block metric on the corrected relaxation gives 201 (star) and 371 (chain). In another row of the same tables ($n = 15$) it compared dense matrix entries with sparse free moments. Version 1 also reported 1.85 s for the solve, on a smaller problem: 1 251 free moments against 3 051 now (star budget).

6.4 Experiment B: estimation error on the kurtosis coefficients

The ball formulation minimises $f(x;\hat{\kappa}) + \delta\,\|Wz(x)\|_2$, where $z(x)$ stacks the monomials x_i^4 and $x_i^2 x_j^2$ and W weights the off-diagonal entries by $\sqrt{2}$. It is the worst case over a Frobenius ball of radius δ around $\hat{\kappa}$. It is parametric robustness; it is not the moment-uncertainty set of Delage and Ye [2]. I draw four markets ($n = 6, 10$; mixed and stress regimes). I perturb the true coefficient vector θ with an error ε of fixed relative size $\rho = \|\varepsilon\|_W / \|\theta\|_W \in \{0.05, 0.1, 0.2, 0.5\}$, with 200 draws per market and ρ: at $\rho = 0.5$ the estimate is off by half the size of the truth. I compare four methods. The nominal problem has $\delta = 0$. The ball uses $\delta = \gamma\|\theta\|_W$ for six values of γ. The automatic rule sets $\delta = c\hat{\sigma}$ from a bootstrap of $T = 20$ synthetic observations. Shrinkage with intensity $s \in [0, 1]$ replaces $\hat{\kappa}$ by $(1-s)\hat{\kappa} + s\,\kappa_{\text{target}}$, where the target has one diagonal value and one off-diagonal value, each the mean of the estimates. The metric is the true regret $f(\hat{x};\kappa) - f(x^\star;\kappa)$:

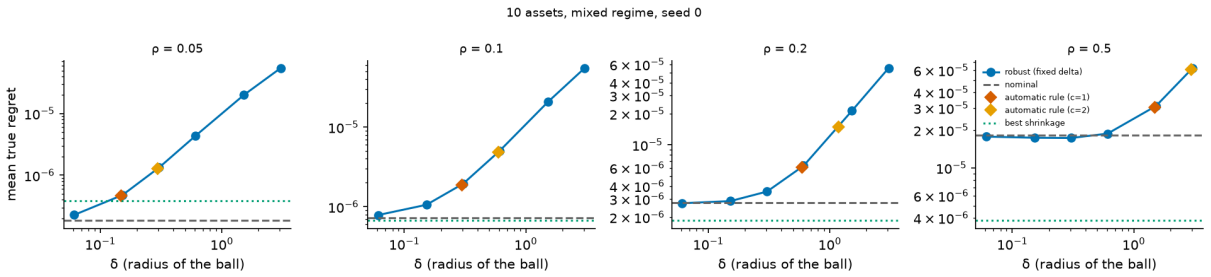


Figure 3: Experiment B, 10-asset mixed market. Mean true regret against δ for the ball (circles), the automatic rule (diamonds), the nominal problem (dashed) and the best shrinkage (dotted), for four noise levels ρ. Log axes.

the extra risk, in the units of f, that the portfolio chosen with the noisy coefficients carries when judged with the true ones. It is never negative; smaller is better, and a ratio to the nominal regret below 1 means the method helps.

Expected. I expected the ball to lower the regret when ρ is large and to cost little when ρ is small.

Observed. On the 10-asset mixed market the ball is worse than nominal for $\rho \le 0.2$, and the loss grows with δ (Figure 3, Table 5). At $\rho = 0.05$ the mean regret is 1.8×10^{-7} for nominal and 1.3×10^{-6} for $\gamma = 0.1$, that is, the ball multiplies the regret by 7.3.

At $\rho = 0.5$ the ball with $\gamma \le 0.1$ lowers the mean regret by 2 to 5%, and the bootstrap intervals exclude zero. At $\gamma = 0.2$ the difference is not distinguishable from zero, and with $\gamma \ge 0.5$ the ball is worse again. Shrinkage with $s = 0.75$ brings the regret to 0.21 of the nominal value. The automatic rule never beats nominal.

In the stress markets the nominal solution is a corner that coincides with $x^\star$ in at least 95% of the draws. One cell discriminates: 10 assets at $\rho = 0.5$, where nominal regret is 2.2×10^{-2}. There the ball with $\gamma = 1$ raises it to 0.72, and shrinkage with $s = 0.5$ brings it to 0.000.

An order-2 SDP check on 1120 solves certifies the nominal local result in 100% of the cases. It certifies the ball results in 99% of the cases for $\gamma = 0.02$ and 25% for $\gamma = 1$.

Reading. The ball adds a penalty proportional to δ, and here the nominal regret is already small. Shrinkage helps more, probably because the synthetic coefficients are drawn i.i.d. from narrow ranges, which makes the mean-based target close to the truth. I did not test this explanation, and the markets are synthetic.

6.5 Experiment C: real data

The data are Yahoo Finance adjusted closes for 37 large US stocks in 9 sectors of the GICS classification, 2009–2025, fixed ex post. The list is today's large caps, so stocks that fell out of the index are absent: this is survivorship bias, and it flatters every strategy. The study is walk-forward. On the first trading day of each month from 2015 to 2025 (132 dates) each strategy estimates its inputs on the previous 756 trading days, about three years, and sets weights. The weights then drift until the next date. Weights are long-only with a cap of 0.25, so that no stock exceeds a quarter of the portfolio; the cap was chosen for the conditioning of the SDP, not for finance (Appendix D). It applies to the minimum-variance, minimum-CVaR and polynomial strategies; 1/N, ERC and HRP are not capped. Costs are 5 or 10 basis points (1 bp $= 10^{-4}$) per unit of traded notional; Appendix E defines turnover. Universe U10 has ten stocks, the first two of each of the first five sectors, and the full (dense) polynomial model; the protocol calls this variant V1. Universe U37 has all 37 stocks and the sector model, which keeps only co-moments between stocks of the same sector (variant V2). In the model $F = \ell_2 m_2 - \ell_3 m_3 + \ell_4 m_4$ of Section 3, the scale $\ell_k = w_k / \overline{|m_k(e_i)|}$ makes each term worth w_k on a portfolio holding a single stock e_i, on average

over the stocks. I fix $(w_2, w_3) = (1, 1)$ and let $w_4 \in \{0, 1, 3\}$ be the weight of the fourth moment; $w_4 = 1$ is the reference value fixed before the runs. Returns are first rescaled so that the average stock volatility is one. The protocol was frozen before any run, with three amendments made before the evaluation period (Appendix D). Indicators are defined once in Appendix E; the rule for reading each is given where the result appears. Comparators are equal weights, Ledoit–Wolf minimum variance [9], equal risk contribution (ERC), hierarchical risk parity (HRP) [10] and minimum CVaR [12]; Appendix F says what each optimises.

Expected. Third and fourth sample co-moments are noisy on 756 days. I expected the global and local minimisers to coincide most of the time, and neither to be distinguishable from simple baselines [3].

Observed. On U10 the certified minimiser equals the local one on every date. On U37 the local solver is suboptimal by more than 10^{-4} in relative terms on 14%, 6% and 4% of the dates, for $w_4 = 0, 1, 3$. Every SOS solve is certified.

Out of sample the picture is flat (Table 6, Figure 4). The net return of a strategy is its mean daily return after costs, times 252; higher is better. Across the 14 strategies of U37 it runs from 8.2% to 14.1% a year, and equal weights earn 12.8%. A difference between two strategies comes with a 95% interval from a block bootstrap (Appendix G); if the interval contains zero, the data do not separate the two. All 13 net-return intervals contain zero. For U10 against equal weights, the net-return difference is -3.1 points a year, with interval -7.5 to $+1.4$, on a base of 15.5%. The Sharpe-ratio test gives p between 0.32 and 0.97 (Appendix C, uncorrected); a p above 0.05 means that a difference this large would arise by chance more than one time in twenty if the two Sharpe ratios were equal.

On risk, the picture depends on the baseline. Volatility is the annualised standard deviation of daily returns, and CVaR is the mean loss on the worst 5% of days; for both, lower is better. Equal weights have a volatility of 16.4% on U10, so a difference of one point is about 6% of the base. With $w_4 = 1$ the U10 model has lower volatility than equal weights, by -1.0 points (interval -2.0 to -0.1). It has higher volatility and CVaR than minimum variance and minimum CVaR.

On U37 the sector model is not better than the local solution of the same model. Its volatility differs by $+0.19$ points ($+0.06$ to $+0.41$), so it is slightly higher.

The price of sparsity is large in model terms. The full-model objective at the sector solution is 0.42 on average against 0.22 at the full-model local solution ($w_4 = 1$). It is worse on 100% of the dates. The objective is dimensionless (each term is scaled by ℓ_k), so only the ratio between the two means is informative.

Time is the wall-clock time per monthly rebalance. The SOS solve takes 25 s per rebalance on U10 and 55 s on U37, against 27 s for the local solve of the sector model. These are wall times with six concurrent workers and are indicative only.

Reading. The sparse global minimiser is better at the thing it optimises, and the gain does not transfer. The most plausible reason is estimation error in the moments, which is what the protocol anticipated. I did not separate it from model error.

7 Discussion and limits

What the experiments show. On synthetic polynomials, local search fails often enough that a certificate has value, especially above ten assets. The order-2 relaxation produces it in most cases, with a cost that exceeds 100 local runs at $n = 50$. On real data the certificate changes the allocation on few dates and does not improve on the risk-based baselines.

What they do not show. They do not show that the certificate is useless for other objectives. They do not show that the ball formulation fails for other uncertainty models, or with real estimation error instead of drawn error. They do not show anything about markets beyond 37 fixed US stocks over one period. The extrapolated dense costs at $n = 30$ and $n = 50$ are not

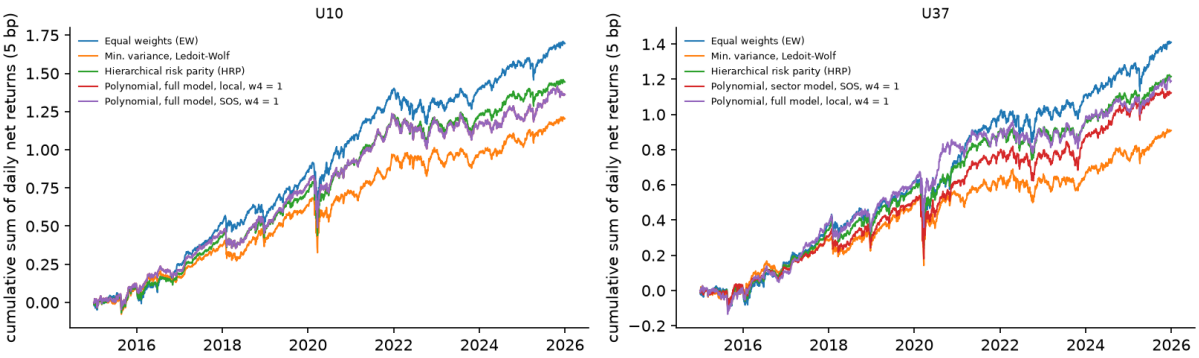


Figure 4: Experiment C. Cumulative sum of daily net simple returns, after costs of 5 bp, for selected strategies, U10 (left) and U37 (right). The sum is not compounded.

measurements.

The bound-accuracy test covers validity and tightness at the 10^{-8} level on small non-convex instances with two or three assets. It does not cover four assets or more, sparse relaxations with several cliques and auxiliary budget variables, or the real-data polynomials. For those I rely on the re-solve sample of Appendix B, which has no independent optimum, so a bias common to all solvers would go undetected.

Explanations I tested: the budget decoupling (fixed and confirmed by the regression test), and the solver accuracy behind the bound excess (Appendix B). Explanations I did not test: why local minima multiply with n, why shrinkage beats the ball in Experiment B, and why the global minimiser does not transfer out of sample. The synthetic polynomial has no return term and a simplified kurtosis structure, so Experiments A and B describe this family only.

Note on tools. The idea, the model and the link between higher-moment portfolio risk and polynomial optimisation are mine. The code was written with an AI assistant, the search for results was AI-assisted, and the text is partly handwritten and partly AI-generated. I reviewed the results, ran an audit of the claims, and corrected the errors it found (Appendix A and `docs/CLAIMS.md`).

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A Corrections to version 1

Table 7 lists the claims of version 1 that did not survive, with the measured values. The register `docs/CLAIMS.md` gives the script and result file behind each line.

B Accuracy of the bound

In Experiment C the bound exceeds the value of the same polynomial at the retained point, by a relative amount up to 1.2×10^{-4}. The sign is always the same, which first looked like an invalid relaxation. I re-solved 39 windows with CLARABEL tolerance 10^{-9} instead of 10^{-7}. Where the solver converges (no weight on the fourth moment, and the 2020 windows of U37) the excess falls by a factor 30 to 100. Where it stalls with an inaccurate status (most cases with $w_4 \geq 1$) the median ratio of the two excesses is 1.00, and the excess stays between 10^{-7} and 1.5×10^{-6} in most cases. The largest excess at tolerance 10^{-9} is 3.5×10^{-6} and none is negative. The polynomial is the same on both sides (checked to nine digits) and the retained points are feasible (largest violation 4×10^{-16}). The reported value is the primal objective of the moment problem; the guaranteed bound would be the dual objective, which I did not extract. On a two-asset instance with a known optimum the excess is 3.2×10^{-9} and the solver residuals are about 10^{-9}, at tolerance 10^{-7} and 10^{-10} alike. I conclude that the excess is a precision floor of the solver, consistent with an unfinished solve, and not a flaw in the relaxation.

Sensitivity of the results. I re-solved a fixed sample of 89 cases (seed 2026) with CLARABEL at 10^{-10} and SCS at 10^{-8} (Table 8). The sample includes inaccurate statuses, uncertified instances and the tightest certificate gaps of each experiment. Across the 161 re-solves, 0 certification verdicts change and 0 failure flags of Experiment A change. The SOS allocations of Experiment C move by at most 7.4×10^{-7}.

With CLARABEL, the bound itself changes by up to 2.6×10^{-5} (A), 4.5×10^{-7} (B) and 1.2×10^{-4} (C, the 2020 windows of U37). SCS differs by up to 8.8×10^{-4} on the 4-asset instances of A, where it also reports an inaccurate status. Its flatness decisions differ from CLARABEL's in 8 of 24 cases of C, so I do not use it for rank decisions. The 5 instances where the sparse bound

is more than 10^{-3} below the dense one stay so with CLARABEL at 10^{-10}. The difference lies between 1.2×10^{-3} and 9.0×10^{-3} and changes by at most 1.1×10^{-5}, so the weaker sparse bound is not a solver artefact.

I treat a change as harmless when it cannot alter a verdict or a figure of the paper. Verdicts use a relative tolerance of 10^{-4}. The quantities that carry conclusions (verdicts, failure flags, allocations, dense-minus-sparse differences) change by at most 1.1×10^{-5} or not at all. The bound values change by more than that on some cases, up to 1.2×10^{-4}, without altering any verdict. I therefore did not rerun Experiments A, B and C.

Independent optimum. A test (`tests/test_bound_soundness.py`) compares the bound with a minimum computed without the SDP on 6 instances with two or three assets, by a fine grid followed by local refinement. It also checks that the bound stays below f at 2000 random feasible points, at the local solution and at the extracted points. At tolerance 10^{-10} the difference between the bound and the independent minimum lies between -8.0×10^{-9} and 3.2×10^{-9}. A tolerance of 10^{-9} cannot be enforced because the solver stalls with residuals of that order. The test therefore allows $+10^{-8}$ (bound above the optimum) and -5×10^{-8} (bound far below it).

C Experiment C: all comparisons

D Reproduction

Software: Python 3.12.3, CVXPY 1.9.3, CLARABEL 0.11.1, NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.6. Machine: Intel Core i5-10310U CPU @ 1.70GHz, 8 threads; experiments used six single-threaded workers. Exact versions are pinned in `requirements.lock`; the package installs with `pip install -no-deps -e .` and the fast tests run with `pytest -m "not slow"`. Experiment protocols are in `experiments/PROTOCOL*.md`, written before the runs. Protocol C has three amendments, all made before the evaluation period: a weight cap of 0.25 in place of 1 for conditioning, an asset-level normalisation of the moment weights, and a cap of 64 on atom combinations in the extraction (disjoint sector cliques otherwise gave about 2×10^6 candidates). Experiments: `experiments/exp_a_local_vs_global.py`, `exp_b_robust.py`, `exp_c_real_data.py`, each followed by its `summarize_*.py`. The prices come from `scripts/download_data.py`. All numbers and tables of this paper are generated by `scripts/make_paper_numbers.py` from `results/`. Experiment C took about 2.0 hours on six workers (the sum of the strategy times in `results/exp_c_alloc.jsonl`, divided by six).

E Indicators

Table 10 gives each indicator as implemented in `empirical.py` and `summarize_exp_c.py`. g_t is the gross daily simple return of the portfolio after drift, and n_t the net return, g_t minus the cost of the trades made that day. The orders of magnitude are the 14 strategies of U37 in Table 6, over 2 765 evaluation days.

Toy example. Table 11 has three assets and five days. Weights are 1/3 and are reset every day, and the cost is $c = 10$ bp. The script `scripts/toy_example.py` computes the numbers below with the code of the experiments, and `tests/test_metric_definitions.py` checks them against values worked out by hand to 10^{-12}. On day 2 the gross return is $(-2-4+3)/3 = -1\%$. After day 1 the portfolio has drifted; going back to 1/3 trades $\sum_i |1/3 - \frac{1}{3}(1+r_i)/(1+g)| = \frac{1}{3}(0.04)/1.02 = 0.0131$ of wealth on day 2, which costs 0.001×0.0131. The mean of the five gross returns is 0.4%, their variance with $T-1 = 4$ is 8.3×10^{-4} and their standard deviation 2.88%. The annualised volatility is $2.88\% \times \sqrt{252}$, which is 45.7% (the toy returns are far larger than real ones). The 5% quantile

of five values lies one fifth of the way from the worst to the second worst, at -3.4%. Only the worst day, -4%, is below it, so the CVaR is 4.0%: with five days the 5% tail is the worst day. The cumulative sum of net returns peaks on day 3 and is lower by 0.0400 on day 4, which is the maximum drawdown. The skewness is -0.52, the annualised net return 1.004 and the Sharpe ratio 2.20. The mean turnover over the four later days is 0.0183. With weights $(0.5, 0.25, 0.25)$ the effective number of assets is $1/0.375 = 2.67$, against 3 for equal weights.

F Comparison strategies

All five use returns only, as the polynomial does, and none forecasts the mean. They span the range from no estimation to tail-risk optimisation.

Equal weights (1/N). Each of the n stocks gets $1/n$. It optimises nothing and assumes nothing about the moments. It is the reference of DeMiguel et al. [3]: an estimated rule that cannot beat it has no use for its estimates.

Minimum variance with Ledoit–Wolf. It minimises $x^\top \hat{\Sigma} x$ on the long-only budget set with the cap, where $\hat{\Sigma}$ is the Ledoit–Wolf estimate. Ledoit–Wolf shrinkage averages the sample covariance with a multiple of the identity, with an intensity computed from the data to cut estimation error [9]. It assumes that variance is the risk. It is the reference for the second-moment part of the polynomial.

Equal risk contribution (ERC). It sets weights so that each stock contributes the same share of portfolio variance, $x_i(\hat{\Sigma}x)_i$ equal across i. The code solves $\min \ \frac{1}{2} y^\top \hat{\Sigma} y - \frac{1}{n}\sum_i \log y_i$ and normalises y. It lies between equal weights and minimum variance, and it has no cap.

Hierarchical risk parity (HRP). It follows López de Prado [10]. It clusters the stocks on the distance $\sqrt{(1-\mathrm{corr}_{ij})/2}$ with single linkage, orders them by the dendrogram, then splits the list in two and gives each half a share inversely proportional to its variance, recursively. It needs no matrix inversion, uses the sample covariance and has no cap. It serves as a reference that is robust to estimation error.

Minimum CVaR. It minimises the mean loss on the worst 5% of the 756 days of the window, as the linear program of Rockafellar and Uryasev [12]: minimise $z + \frac{1}{(1-0.95)T}\sum_t u_t$ with $u_t \geq -r_t^\top x - z$, $u_t \geq 0$, on the capped budget set. It uses the empirical distribution and no model. It is the closest competitor to a tail-risk polynomial, since both target the tails.

G Vocabulary

Statistics. *In-sample* is the window used to estimate; *out-of-sample* is the later period on which the result is judged. A *backtest* replays a strategy on past data. *Look-ahead* is the use of information not available at the decision date; here each window ends the day before the rebalance. *Overfitting* is a rule that fits the noise of its estimation window. *Survivorship bias* is the absence of stocks that failed. A *rolling window* moves forward with the decision date. The *95% interval* of a difference is the range from the 2.5th to the 97.5th percentile of the difference over 2000 resamples of the evaluation days. The resampling is a *circular block bootstrap*: it draws blocks of 21 consecutive days, to keep the dependence between neighbouring days. A *p-value* is the probability, if the null hypothesis were true, of a statistic at least as extreme as the one observed; the Sharpe test uses blocks of 5 days and 2000 resamples. With *multiple comparisons*, the chance that some interval

excludes zero grows with their number. For 13 independent comparisons at 5% one expects 0.65 such intervals by chance alone; the comparisons here are dependent, and no correction is applied.

Optimisation. A *local minimum* is a point no feasible neighbour improves; a *global minimum* is the best feasible point. A local solver stops at the first local minimum it reaches. *Multi-start* runs it from many random points and keeps the best; it gives no proof. A *relaxation* replaces the problem by an easier one whose feasible set is larger, so its optimum λ_{lb} is a lower *bound*: no feasible point has a smaller value. A *certificate* is a feasible point $\hat{x}$ with $f(\hat{x}) - \lambda_{\mathrm{lb}} \leq \max(10^{-6}, 10^{-4}|\lambda_{\mathrm{lb}}|)$. Then $\hat{x}$ is ε-optimal: within ε of the global minimum, ε being that tolerance. The *solver tolerance* is the accuracy at which an interior-point method stops; such a method follows a path through the interior of the feasible cone. "Certified" therefore means certified to the precision of the solver (Appendix B), not exactly. *SDP* (semidefinite program) is optimisation over matrices constrained to be positive semidefinite.

Table 2: Expectations against measurements. "Held" means the measurement agrees with the expectation on the tested instances.

Expectation	Reason	Measurement	Verdict	Explanation
Sparse bound equals dense bound	Theorem 7 of version 1	Version 1: -22.1 against -2.43 on one instance. After the budget fix, median dense minus sparse 2.9×10^{-10}, but 5 of 1100 instances exceed 10^{-3} (max 9.0×10^{-3})	Partly	Budget bug in version 1; remaining gaps are on instances the sparse relaxation does not certify
2-asset benchmark is non-convex	Version 1 caption	Smallest Hessian eigenvalue on $\mathcal{X}$: $+1.68$; one minimum	Not held	The benchmark is convex; a non-convex instance has to be built (Sec. 6.1)
Local search is enough in the default regime	Protocol of Exp. A	Several local minima in 48% of default-regime instances at $n = 15$ and 98% at $n = 50$; one start fails with probability 0.30 at $n = 15$ and 0.83 at $n = 50$	Not held	The number of local minima grows with n; I did not test why
Order-2 certificate in most cases	Tightness of low orders	All but 29 of 1760 instances certified; 74% for the default regime at $n = 50$	Held, except at large n	79 solves return an inaccurate status
Flat extension at $d = 2$ certifies	Version 1	Flat in all but 46 instances; at $d = 3$ on the non-convex pair the ranks are 5 and 1 with the same bound	Unstable	The solver may return a point inside the optimal face; not tested
Size reduction of 398 at $n = 50$	Version 1 table	128 and 215 (PSD entries), 104 and 135 (moments), for star and chain	Corrected	Version 1 counted moment-matrix entries only
Dense SDP unaffordable at $n = 50$	"$>$ 60 h" in version 1	Measured: 193 s and 4.7 GB at $n = 15$. Extrapolated, not measured: 3.8×10^6 s at $n = 50$	Not tested	Extrapolation from five points
SDP faster than local search	Cost of one solve	At $n = 50$ in the default regime, 80 s against 33 s for 100 local runs	Not held	The sparse SDP still has blocks of size up to 66
Ball formulation helps under kurtosis error	Experiment B design	Worse than nominal for $\rho \leq 0.2$; 5% better at $\rho = 0.5$ with $\gamma = 0.1$	Not held	The ball adds a bias proportional to δ; not otherwise tested
Certified minimiser better out of sample	Motivation	Net-return intervals contain zero in all 13 comparisons; volatility and CVaR are above those of minimum variance and minimum CVaR	Not held	Moment estimation error, not separated from model error; Sec. 6.5

Table 3: Experiment A, selected cells. Multi-min: share of instances with several distinct local minima among 100 runs; p_1: mean probability that one start misses the reference optimum; fail K: share of instances where the best of K starts misses it ("cert.": among the instances whose reference is the certified point); uncert.: share of instances that are not certified at $d = 2$, where the reference is the best of 100 runs and the 100-start failure is false by construction; times are medians of the sparse SDP (star budget, build included) and of 100 local runs, in seconds, with six concurrent workers. The full grid is in `results/exp_a_summary.md`.

regime	n	inst.	multi-min	p_1	fail 1	fail 100	fail 100 cert.	uncert.	SDP (s)	100 local (s)
default	4	50	0%	0.00	0%	0.0%	0.0%	0%	0.1	0.7
default	10	50	18%	0.09	6%	0.0%	0.0%	0%	0.6	1.4
default	15	50	48%	0.30	32%	0.0%	0.0%	0%	0.7	1.9
default	30	50	92%	0.65	60%	0.0%	0.0%	0%	2.3	3.9
default	50	50	98%	0.83	86%	16.0%	21.6%	26%	79.9	32.7
mixed	4	50	52%	0.23	20%	0.0%	0.0%	14%	0.1	0.5
mixed	10	50	84%	0.52	60%	0.0%	0.0%	0%	0.5	0.8
mixed	15	50	92%	0.66	52%	0.0%	0.0%	6%	0.6	0.9
mixed	30	50	100%	0.77	66%	0.0%	0.0%	4%	1.9	1.7
mixed	50	50	100%	0.86	82%	0.0%	0.0%	2%	67.4	11.9
stress	4	120	65%	0.29	22%	0.0%	0.0%	1%	0.1	0.4
stress	10	120	80%	0.59	74%	0.0%	0.0%	0%	0.5	0.6
stress	15	120	85%	0.69	73%	0.0%	0.0%	0%	0.6	0.7
stress	30	120	100%	0.85	72%	0.0%	0.0%	0%	1.8	1.1
stress	50	120	100%	0.90	90%	0.0%	0.0%	0%	19.7	2.1

Table 4: Dense versus sparse relaxation, order $d = 2$, one synthetic instance per n. Dense times and memory are measured up to $n = 15$; rows marked * are log-log extrapolations from five measured points, not measurements (95% prediction intervals in `results/dense_extrapolation.csv`). Sparse figures use the chain budget.

n	dense block	dense (s)	dense (GB)	dense status	sparse block	sparse (s)
4	15	0.07	0.12	optimal	6	0.07
6	28	0.21	0.13	optimal	10	0.09
8	45	1.62	0.19	optimal	15	0.13
10	66	6.12	0.39	optimal	21	0.50
12	91	40.62	1.06	optimal inaccurate	15	0.26
15	136	192.81	4.67	optimal inaccurate	21	0.71
30	496	$5.3 \times 10^{4*}$	58^*	extrapolated	21	6.6
50	1326	$3.8 \times 10^{6*}$	546^*	extrapolated	21	26.3

Table 5: Experiment B, mean true regret over 200 draws, mixed markets. Nominal: $\delta = 0$; ball: $\delta = \gamma\|\theta\|_W$ with $\gamma = 0.1$ and 1; rule: $\delta = c\,\hat{\sigma}$ with $c = 1$; shrinkage towards the structured target with $s = 0.5$ and 1.

n	ρ	nominal	ball 0.1	ball 1	rule	shrink 0.5	shrink 1
10	0.05	1.8×10^{-7}	1.3×10^{-6}	5.5×10^{-5}	4.6×10^{-7}	1.1×10^{-6}	4.5×10^{-6}
10	0.2	2.8×10^{-6}	3.5×10^{-6}	5.5×10^{-5}	6.1×10^{-6}	1.8×10^{-6}	4.5×10^{-6}
10	0.5	1.8×10^{-5}	1.7×10^{-5}	6.2×10^{-5}	3.0×10^{-5}	5.7×10^{-6}	4.8×10^{-6}
6	0.05	0	1.0×10^{-3}	5.9×10^{-1}	0	0	0
6	0.2	3.5×10^{-4}	3.3×10^{-2}	5.9×10^{-1}	1.5×10^{-1}	9.6×10^{-5}	4.0×10^{-5}
6	0.5	7.4×10^{-2}	1.2×10^{-1}	5.9×10^{-1}	4.4×10^{-1}	3.0×10^{-2}	1.8×10^{-2}

Table 6: Experiment C, out of sample 2015–2025 (132 monthly rebalances). Annualised volatility, daily 5% CVaR, maximum drawdown (%), mean turnover per rebalance ($\sum_i |w_i - w_i^{\text{drift}}|$, buys and sells counted), effective number of assets, annualised return net of 5 bp, Sharpe ratio net of 5 bp. $w_4 = 1$ for the polynomial models.

universe	strategy	vol (%)	CVaR (%)	MDD	turn.	N_{eff}	net (%)	Sharpe
U10	EW	16.4	2.43	36	0.04	10.0	15.5	0.94
U10	MinVar_LW	14.7	2.11	36	0.05	5.5	11.0	0.75
U10	ERC_LW	15.7	2.31	36	0.04	9.6	14.3	0.91
U10	HRP	15.3	2.24	35	0.11	8.1	13.2	0.86
U10	MinCVaR95	14.8	2.13	35	0.10	5.1	10.9	0.74
U10	Poly_full_local	15.4	2.23	34	0.14	4.6	12.4	0.81
U10	Poly_full_SOS	15.4	2.23	34	0.14	4.6	12.4	0.81
U37	EW	16.7	2.47	40	0.04	37.0	12.8	0.77
U37	MinVar_LW	14.4	2.06	40	0.12	9.3	8.3	0.58
U37	ERC_LW	15.7	2.32	38	0.04	34.4	11.9	0.76
U37	HRP	15.0	2.21	35	0.15	29.0	11.1	0.74
U37	MinCVaR95	14.6	2.05	40	0.23	7.2	8.2	0.56
U37	Poly_full_local	14.9	2.13	32	0.27	5.9	10.8	0.73
U37	Poly_sector_local	14.7	2.14	36	0.07	18.6	10.1	0.69
U37	Poly_sector_SOS	14.9	2.17	36	0.11	17.9	10.2	0.69

Table 7: Claims of version 1 and their status.

Claim in version 1	Measured	Status
2-asset benchmark is non-convex	Smallest Hessian eigenvalue on $\mathcal{X}$: +1.68; one local minimum	Refuted
Default instances show the value of global search	Sparse bound, dense bound and local value coincide on the five default instances; one local minimum each	Refuted
Sparse bound equals dense bound	Instance $(6, 2, 0)$, skew 2: -22.1 against -2.43; instance $(6, 2, 1)$, skew 4: -174 against -5.4. Cause: budget imposed on $E[\sum x]$	Refuted, fixed
398-fold reduction at $n = 50$	$399 = 1\,758\,276/4\,410$ compares the entries of the dense moment matrix with those of the ten clique moment matrices, without localising matrices or budget clique; free moments give 253; the row $n = 15$ mixed matrix entries with free moments	Corrected
Dense solve > 60 s ($n = 15$) and > 60 h ($n = 50$)	Not SDP timings: they came from the scalar-loop builder	Unsupported
Extraction for rank above one	Random combination with non-orthogonal eigenvectors, atoms merged by position	Rewritten
Local solver returns a feasible bound	Scaling projection, success flag always true	Rewritten
"Distributionally robust (Delage–Ye)"	Frobenius ball on the kurtosis coefficients	Renamed
Guarantee "inaccessible to gradient methods"; real-time rebalancing	Promotional or unsupported	Removed

Table 8: Re-solves of a fixed sample (seed 2026) with CLARABEL at 10^{-10} and SCS at 10^{-8}, against the stored CLARABEL 10^{-7} results. Largest relative change of the bound, relative to $\max(1, |\text{lb}|)$ in B and C and to $\max(10^{-3}, |\text{lb}|)$ in A. Flips: certification verdicts that change.

exp.	cases	CLARABEL 10^{-10}	SCS 10^{-8}	verdict flips	other
A	30	2.6×10^{-5}	8.8×10^{-4}	0 of 48	failure flags changed: 0
B	30	4.5×10^{-7}	4.7×10^{-7}	0 of 60	certification decision only
C	29	1.2×10^{-4}	1.3×10^{-5}	0 of 53	largest allocation change 7.4×10^{-7}

Table 9: Experiment C, all 13 pairwise comparisons, difference A minus B in percentage points (annualised volatility; daily 5% CVaR; annualised net return at 5 bp) with 95% circular-block bootstrap intervals (blocks of 21 days), and the p-value of the Ledoit–Wolf Sharpe-ratio test. No multiple-comparison correction; the intervals are descriptive. On U10 the two solutions of the full model coincide on every date, so the row "SOS full" against "local full" compares two identical series and its p-value is numerical noise.

U	A	B	vol	CVaR	net return	p
U10	SOS full	EW	−1.00 [−1.96, −0.14]	−0.20 [−0.32, −0.08]	−3.1 [−7.5, +1.4]	0.33
U10	SOS full	MinVar	+0.72 [+0.25, +1.20]	+0.12 [+0.02, +0.21]	+1.4 [−1.4, +4.2]	0.60
U10	SOS full	ERC	−0.28 [−1.10, +0.47]	−0.08 [−0.19, +0.03]	−1.9 [−5.5, +1.8]	0.38
U10	SOS full	HRP	+0.06 [−0.64, +0.72]	−0.02 [−0.11, +0.08]	−0.8 [−3.8, +2.3]	0.62
U10	SOS full	MinCVaR	+0.58 [+0.27, +0.92]	+0.10 [+0.03, +0.17]	+1.5 [−1.1, +4.1]	0.49
U10	SOS full	local full	+0.00 [−0.00, +0.00]	+0.00 [−0.00, +0.00]	+0.0 [−0.0, +0.0]	0.71
U37	SOS sector	EW	−1.80 [−2.22, −1.36]	−0.29 [−0.39, −0.19]	−2.6 [−5.9, +0.7]	0.43
U37	SOS sector	MinVar	+0.48 [−0.06, +1.05]	+0.11 [+0.01, +0.20]	+1.9 [−1.0, +5.2]	0.32
U37	SOS sector	ERC	−0.84 [−1.15, −0.45]	−0.15 [−0.22, −0.06]	−1.7 [−4.0, +0.6]	0.38
U37	SOS sector	HRP	−0.18 [−0.40, +0.09]	−0.03 [−0.09, +0.03]	−0.9 [−2.6, +1.2]	0.49
U37	SOS sector	MinCVaR	+0.31 [−0.33, +0.97]	+0.12 [+0.01, +0.22]	+2.0 [−1.4, +5.6]	0.33
U37	SOS sector	local sector	+0.19 [+0.06, +0.41]	+0.03 [−0.00, +0.08]	+0.1 [−1.2, +1.5]	0.97
U37	SOS sector	local full	−0.01 [−0.52, +0.45]	+0.05 [−0.05, +0.16]	−0.6 [−4.2, +2.9]	0.76

Table 10: Indicators: formula as implemented, unit and direction, measured range, limit of interpretation.

Indicator	Formula	Unit; better is	Measured (U37)	Limit
Annualised volatility	$\sqrt{252}\,\text{sd}(g_t)$, sd with $T-1$	fraction a year (% in tables); lower	14.4–16.7%; equal weights 16.7%	gains and losses count alike; $\sqrt{252}$ assumes independent days
CVaR at 5%	$-\overline{g_t}$ over the days with $g_t \le q$; q is the 5% quantile of g (linear interpolation)	daily loss, positive; lower	2.05–2.47% a day	rests on about 138 days out of 2 765
Maximum drawdown	$\max_t \left(\max_{u \le t} C_u - C_t\right)$ with $C_t = \sum_{u \le t} n_t$	fraction of initial wealth; lower	28.8–40.5%	sums returns, does not compound; measured from the peak in return units, not in % of the peak; one path
Skewness	$\frac{1}{T}\sum_t z_t^3$, $z_t = (g_t - \bar{g})/\text{sd}(g)$	none; higher	−0.52 to −0.18	a few days dominate it
Turnover	mean over rebalances after the first of $\sum_i \lvert x_i - w_i \rvert$, w the drifted weights	share of wealth, buys and sells both counted; lower	0.04–0.38; equal weights 0.04	cost assumed linear in the traded amount; no market impact
Net return	$252\,\overline{n_t}$; $n_t = g_t - c\sum_i \lvert x_i - w_i \rvert$ on a rebalance day, g_t otherwise; $c = 5$ or 10 bp	fraction a year; higher	8.2–14.1%	the polynomial has no return term, so no strategy here optimises it
Sharpe ratio	$\sqrt{252}\,\bar{n}/\text{sd}(n)$; risk-free rate zero	none; higher	0.56–0.90	ignores the return on cash; one noisy estimate
Effective number of assets	$1/\sum_i x_i^2$, mean over rebalances	between 1 and n; higher is more spread	5.4–37.0	looks at weights, not correlations
Regret (Exp. B)	$f(\hat{x};\kappa) - f(x^\star;\kappa)$	units of f, ≥ 0; lower	1.8×10^{-7} for nominal, 10 assets, $\rho = 0.05$	depends on the scale of f; compare ratios
Single-start failure (Exp. A)	share of the 100 local runs ending more than τ above the reference	probability; lower	0.30 at $n = 15$ (default regime)	depends on how starts are drawn

Table 11: Toy example: three assets, five days, equal weights rebalanced daily, cost 10 bp per unit traded. Returns and gross and net portfolio returns in %; turnover is $\sum_i |w_i - w_i^{\text{drift}}|$ before the rebalance.

day	A	B	C	gross	turnover	net
1	4	2	0	2.00	0.0000	2.0000
2	−2	−4	3	−1.00	0.0131	−1.0013
3	6	3	0	3.00	0.0269	2.9973
4	−6	−3	−3	−4.00	0.0194	−4.0019
5	3	0	3	2.00	0.0139	1.9986